Cover Letter

Date: August 3, 2026

To: The Editors, *Genome Biology*

Re: Submission of "Delta Dependency Profiles Paralog Compensation Signals Across Solid Tumor Types" (Research article)

Dear Editors,

We submit for consideration in *Genome Biology* our manuscript **"Delta Dependency Profiles Paralog Compensation Signals Across Solid Tumor Types."**

Paralog compensation is an attractive route to new cancer dependencies, but most computational synthetic-lethality (SL) predictors are multi-feature models whose nominations are hard to trace to a single measurable effect. We introduce Delta Dependency (DD), the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines: each nomination traces directly to a measured dependency shift and an immediately designable validation experiment.

Three results define the study. First, on a citation-verified, evidence-tiered benchmark of 12 curated paralog-SL pairs, signed DD achieves AUROC = 0.629 in a positive–unlabeled evaluation with explicit label-quality sensitivity analyses (permutation $p = 0.122$; at the current sample size of 8 positive entries, power to reject a null of 0.5 is 34%); an interpretable composite score produced a similar point estimate to the best tested classifier on the same pairs, with wide uncertainty given only six positive pairs (the composite weights were developed against this benchmark and its AUROC is therefore descriptive context, not independent validation). The only lineage-evaluable Tier A/B pair, ARID1A→ARID1B, showed consistently increased dependency in two gynecological lineages; because this represents one biological pair, we do not treat the result as an independent performance estimate. Second, two combinatorial CRISPR screens published in 2025 (Harle et al., *Genome Biology*; Flister et al., *Cell Reports*) served as fully external tests that played no role in method development or candidate selection. In a mutation-agnostic analysis (max|DD|, which does not test the directional, mutation-conditioned proposition that defines signed DD), dependency displacement separates experimentally determined digenic synthetic lethals from non-hits in both datasets (AUROC = 0.619 and 0.696; permutation $p = 0.0002$ and 0.0001), on experimental platforms independent of DepMap. The genotype-stratified, directional test could not be adjudicated in either screen because only 3 curated pairs per dataset had any driver-mutant models (9 and 6 mutant observations respectively). Third, an in vitro dependency-window score recovers ARID1A→ARID1B as the highest-selectivity established dependency (selectivity = +0.28; SMARCA4→SMARCA2 has the higher DWS, 4.87 vs 2.82, but ARID1A→ARID1B has the higher selectivity) and nominates KMT2D→KMT2C as the highest-ranked non-benchmark candidate for combinatorial-CRISPR testing. DD is an association-based prioritization metric, not a causal test of synthetic lethality; all candidates require experimental validation.

We believe this work suits *Genome Biology* because it pairs a mechanistically transparent metric with broad pan-cancer evaluation and an honestly bounded evidence chain. Relative to the recent combinatorial screens, which map paralog buffering experimentally and at scale, our contribution is complementary: a direction-aware, mutation-conditioned prioritization framework whose every quantitative claim is machine-auditable. All data are public; the analysis pipeline (https://github.com/tjogzt/paralog-sl-predictor) and the companion `paralogSL` R package (https://github.com/tjogzt/paralogSL) are openly available under the MIT license and archived on Zenodo (concept DOIs 10.5281/zenodo.21502030 and 10.5281/zenodo.21502113, which always resolve to the latest release; version-specific DOIs, release tag, and commit are listed in the manuscript). A single entry point (`verify_all.sh`) runs four audit modules plus a 31-test suite, recomputing 420 numeric claims directly from the analysis artifacts in under a minute, with all figures generated by scripts from the same artifacts.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have read and approved the final manuscript and agree with its submission to *Genome Biology*. The authors declare no competing interests. Q.M. and T.Z. conceived the study; T.Z. developed the methodology, performed all computational analyses, developed the R package, and wrote the original draft; Q.M. curated clinical datasets, provided gynecological oncology domain expertise, and contributed to writing and revision; P.C., C.X., Y.W., T.H. and Q.S. contributed clinical data curation and clinical interpretation; C.X. and Y.W. additionally acquired funding.

Suggested reviewers (also entered in the submission system):

1. Dr. Francisca Vazquez, DepMap / Broad Institute (CRISPR dependency expertise); vazquez@broadinstitute.org
2. Dr. Jason Moffat, University of Toronto / The Hospital for Sick Children (synthetic-lethality and combinatorial CRISPR screening); jason.moffat@sickkids.ca
3. Dr. Bing Zhang, Baylor College of Medicine (CPTAC proteogenomics); bing.zhang@bcm.edu
4. Dr. G. Traver Hart, Department of Systems Biology, UT MD Anderson Cancer Center (combinatorial CRISPR screens, paralog synthetic lethality); traver@hart-lab.org
5. Dr. Min Wu, Institute for Infocomm Research, A*STAR, Singapore (machine-learning benchmarking for SL prediction); wumin@i2r.a-star.edu.sg

We thank you for your consideration and look forward to your response.

Sincerely,

Tao Zhu (on behalf of all authors)

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